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$$\langle \psi_a | \psi_b \rangle = \sum_i c_{i(a)}^* c_{i(b)} = \begin{pmatrix} c_{1(a)}^* & c_{2(a)}^* & \cdot & \cdot & \cdot \end{pmatrix} \begin{pmatrix} c_{1(b)} \\ c_{2(b)} \\ \vdots \end{pmatrix} = \vec{c}_a^\dagger \vec{c}_b$$

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$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ \vdots \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ \vdots \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ \vdots \end{pmatrix}, \dots$ of an infinite dimensional vector space

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$\downarrow$ $\swarrow$

$$\sqrt{\frac{2}{a}} \sin \frac{\pi}{a} x \quad \sqrt{\frac{2}{a}} \sin \frac{2\pi}{a} x \quad \forall 0 < x < a \quad \text{and } 0 \text{ elsewhere}$$

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we can label them as $|1\rangle, |2\rangle, |3\rangle, \dots$

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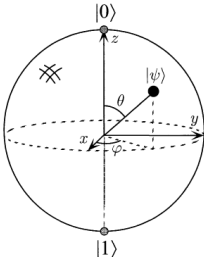
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or: $\begin{pmatrix} \cos \xi \\ e^{i\phi} \sin \xi \end{pmatrix}$; underlying field set of complex numbers, $\{\xi, \phi\} \in \mathbb{R}$

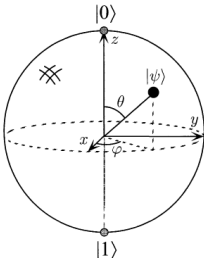
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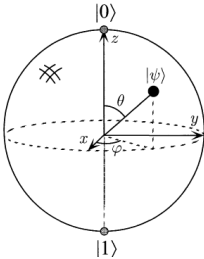


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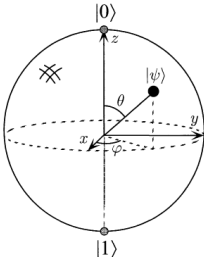
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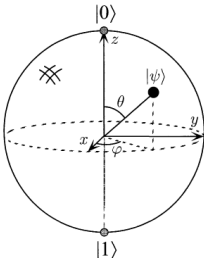


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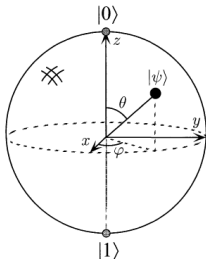


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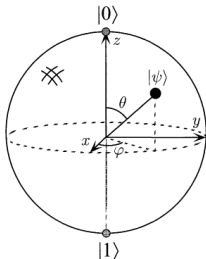


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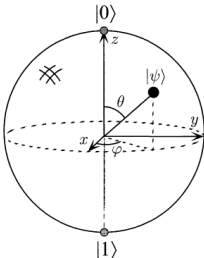
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only if infinitely many identically prepared qubits are measured, one can determine a and b for ψ

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given: $\widehat{S}_+|\alpha\rangle = 0$
 $\widehat{S}_+|\beta\rangle = \hbar|\alpha\rangle$

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Ex. 4: in the basis of energy eigenstates, the Hamiltonian matrix is diagonal

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then show : it is same as sum of squares of the components

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- can be generalised to higher dimensions

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corollary: Operators corresponding to measurable properties must be Hermitian

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then $|\psi\rangle = \hat{A}^{-1}|\psi'\rangle$

It can be easily seen that $\hat{A}^{-1}\hat{A} = \hat{I}$

$$\text{and } \boxed{(\hat{A}\hat{B})^{-1} = \hat{B}^{-1}\hat{A}^{-1}}$$

Matrix A^{-1} corresponding to operator $\hat{A}^{-1}$

is inverse of matrix A corresponding to operator $\hat{A}$

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$$\vec{c}^{(n)} \Rightarrow \begin{pmatrix} c_1^{(n)} \\ c_2^{(n)} \\ . \\ . \\ . \end{pmatrix}; \text{ Normalise: } \sum |c_i^{(n)}|^2 = 1 \implies \vec{c}^{(n)\dagger} \vec{c}^{(n)} = 1$$

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For $a = 1$, $\vec{c} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and for $a = -1$, $\vec{c} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

T2. Eigenfunctions of Hermitian operator corresponding to different eigenvalues
are orthogonal

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and $f(x) = \frac{4l}{\pi^2} \left[\sin\left(\frac{\pi x}{l}\right) - \frac{1}{3^2} \sin\left(\frac{3\pi x}{l}\right) + \dots \right]$

Average value of a physical observable B at time t , $\langle B \rangle = \frac{\int \Psi^* \hat{B} \Psi d\tau}{\int \Psi^* \Psi d\tau} \equiv \frac{\langle \Psi | \hat{B} | \Psi \rangle}{\langle \Psi | \Psi \rangle}$

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For a particular eigenstate,

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To obtain time dependence in any experimental observable

it is necessary to have a superposition of states with different energies

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$$\text{Thus, } \langle p \rangle = m \frac{d\langle x \rangle}{dt} = \left\langle \frac{\hbar}{i} \frac{\partial}{\partial x} \right\rangle \leftarrow p_i \equiv \frac{\hbar}{i} \frac{\partial}{\partial q_i}$$

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$$\therefore \hat{A}\hat{B}g = \hat{B}\hat{A}g \quad \forall g \text{ (important : it is } \forall g \text{ and not just for special case)}$$

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$\therefore \hat{B}g_i$ is an eigenfunction of the operator $\hat{A}$ with same eigenvalue a_i as eigenfunction g_i

i.e., $\hat{B}g_i = k g_i$ where k is some number, or g_i is an eigenfunction of $\hat{B}$

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we can select a common complete set of eigenfunctions for them

Proof : Let g_i be eigenfunction of $\hat{A}$, s.t. $\hat{A}g_i = a_i g_i$.

consider the non-degenerate case when for each eigenvalue there is only one eigenfunction

$$\hat{B}(\hat{A}g_i) = \hat{B}(a_i g_i) = a_i \hat{B}g_i \quad \text{but } \hat{A}\hat{B} = \hat{B}\hat{A}; \quad \therefore \hat{A}(\hat{B}g_i) = a_i (\hat{B}g_i)$$

$\therefore \hat{B}g_i$ is an eigenfunction of the operator $\hat{A}$ with same eigenvalue a_i as eigenfunction g_i

i.e., $\hat{B}g_i = kg_i$ where k is some number, or g_i is an eigenfunction of $\hat{B}$

This is true for all g_i s in the set

T4. If Hermitian operators $\hat{A}$ and $\hat{B}$ commute, we can select a common complete set of eigenfunctions for them

Proof : Let g_i be eigenfunction of $\hat{A}$, s.t. $\hat{A}g_i = a_i g_i$.

consider the non-degenerate case when for each eigenvalue there is only one eigenfunction

$$\hat{B}(\hat{A}g_i) = \hat{B}(a_i g_i) = a_i \hat{B}g_i \quad \text{but } \hat{A}\hat{B} = \hat{B}\hat{A}; \quad \therefore \hat{A}(\hat{B}g_i) = a_i (\hat{B}g_i)$$

$\therefore \hat{B}g_i$ is an eigenfunction of the operator $\hat{A}$ with same eigenvalue a_i as eigenfunction g_i

i.e., $\hat{B}g_i = k g_i$ where k is some number, or g_i is an eigenfunction of $\hat{B}$

This is true for all g_i s in the set

For the degenerate case, the proof is a bit complex and for now, we are not considering it.

Summary of theorems T1-T4:

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Eigenfunctions of a Hermitian operator form a complete, orthonormal set

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Eigenfunctions of a Hermitian operator form a complete, orthonormal set
and the eigenvalues are real (normalisation by choice)

T5. If g_i and g_j are eigenfunctions of the Hermitian operator $\hat{A}$ with different eigenvalues (i. e., $\hat{A}g_i = a_i g_i$ and $\hat{A}g_j = a_j g_j$ with $a_i \neq a_j$)

and if $\hat{B}$ is linear operator that commutes with $\hat{A}$ then $\langle g_j | \hat{B} | g_i \rangle = 0$, for $a_i \neq a_j$